



United International University

Department of Computer Science and Engineering

CSE 4495: Software Testing and Quality Assurance

Final Examination : Fall 2024

Total Marks: 40 Time: 2 hours

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

Answer all the questions. Numbers to the right of the questions denote their marks.

1. (a) *SmartHome Solutions Inc.* has developed a new home automation system that allows users to control devices such as smart lights, thermostats, and security cameras through a mobile app. As a performance tester, you identify the following issues:
- When multiple users in a household try to control devices simultaneously, there is a noticeable delay of several seconds before the system processes commands.
 - During peak hours, the server struggles to handle requests, causing timeouts and occasional crashes.
 - As more users adopt the system, performance degrades significantly, and new devices take a long time to sync with the cloud service.

Which quality attribute(s) are most lacking in this system? Justify your answer with proper reasoning and suggest potential optimizations to improve the system. [4]

- (b) A cloud-based e-commerce platform is undergoing reliability testing before launch. The system must meet the following release requirements: Minimum availability: 99.5% and ROCOF (Rate of Occurrence of Failures) must be below 8 failures per 24-hour period. After 15 days of continuous testing, the following data were recorded.

- Total transactions processed: 12,000
- System failures recorded: 18
- Average downtime per crash (including reboot & recovery time): 25 minutes
- 60% of failed transactions resulted in a system crash

- Calculate *POFOD*, *ROCOF* and *Availability* from the given data. [4]
- What is the *MTTR* of this system? Does the system meet release requirements? [2]

2. A movie streaming platform is testing a new feature which calculates a user's subscription fee based on user preferences such as subscription plan, number of simultaneous streams, and device type. The calculation of subscription fee follows the the given requirements -

R1: The base fee for all subscriptions is USD 10 per month.

R2: Three plan types are available - *i)* Basic Plan has no change to the base fee. *ii)* Standard Plan increases the fee by 50%. *iii)* Premium Plan doubles the base fee (+100%).

R3: The number of simultaneous streams adjusts the fee accordingly: 1 stream has no additional charge, whereas, 2 and 4 streams increase the fee by 10% and 20% respectively.

R4: The device type used for streaming also impacts the fee: *i)* Mobile devices receive a 5% discount. *ii)* TV streaming adds a 10% increase to the fee. *iii)* Laptop/PC streaming has no effect on the base fee.

R5: The maximum fee for any subscription plan is capped at USD 50 per month.

So, you need to write test cases to test if all of these requirements holds or not.

- Identify your choices while testing this feature. Choices are aspects that you have control over that can alter the output of the program. Also enumerate representative values for your choices. [6]
 - Calculate the number of test specifications for your choices. [2]
 - Now assume you want to reduce the number of specifications obtained in the previous question. Explain why *CIT* is applicable in this scenario instead of *CPM*. Use *CIT* to find the 2-way covering array over your set of choices. [4]
3. You are developing a fitness tracking app that helps users track their daily exercise routines. The app allows users to log exercises with their duration and intensity (e.g., Low, Medium, High). If a user logs multiple exercises, they will be sorted in order of intensity (*HIGH* > *MEDIUM* > *LOW*). Exercises with the same intensity will be displayed in the order they were logged. As you have developed the app, now you need to write unit test cases in JUnit notation to test the app in following scenarios-

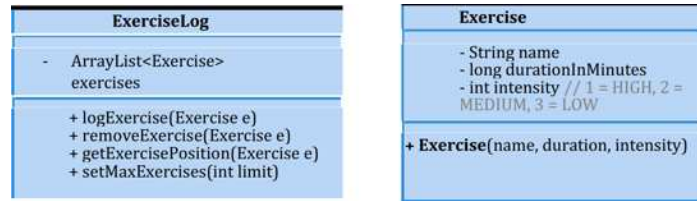


Figure 1: Class diagrams for Ques-3

- (a) Create exercises with varying intensities in the following order: Low, Medium, High, Low, and Medium. After each exercise is logged, check if it was inserted in the correct position based on its intensity using the `getExercisePosition(Exercise e)` method. [4]
 - (b) Write a test case to check that the system enforces a maximum limit on the number of exercises that can be logged for the day. Set the daily limit to 3 exercises. If you attempt to log more than 3 exercises and verify that the system throws a *ScheduleFullException* with the message: “Your daily exercise limit has been reached. Increase the limit to add more exercises.” [4]
4. Consider the following code for a `maxDifference` function that calculates the maximum difference between two numbers in an array, where the larger number comes after the smaller one:

```

1. public int maxDifference(int arr[]) {
2.     if (arr == null || arr.length < 2) {
3.         return -1; // Invalid input, return -1
4.     }
5.     int minElement = arr[0];
6.     int maxDiff = arr[1] - arr[0];
7.
8.     for (int i = 1; i < arr.length; i++) {
9.         if (arr[i] - minElement > maxDiff) {
10.            maxDiff = arr[i] - minElement;
11.        }
12.        if (arr[i] < minElement) {
13.            minElement = arr[i];
14.        }
15.    }
16.    return maxDiff;
17.}

```

- (a) Draw the Control Flow Graph (CFG) for the `maxDifference` method. [4]
- (b) Consider the two test cases :
 - TC1 - `maxDifference([5,10])`
 - TC2 - `maxDifference([10, 8, 7, 6, 4])`

Trace the execution flow of these cases on your CFG. After executing each test calculate the statement and branch coverage of the test suite. [4]
- (c) Check whether or not your test suite has achieved *Loop Boundary Coverage*. If not add more tests to achieve coverage [2]